llmXive Automated Review of MolmoMotion: Forecasting Point Trajectories in 3D with Language Instruction

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision
The review focuses on the accuracy of factual claims and the support provided by citations and data within the manuscript.

1. Unsupported Comparative Claim (Section 3, Method): In the "2D point tracking and metric 3D lifting" subsection, the authors state: "We empirically find that this paradigm [ViPE] produces more accurate 3D trajectories than current end-to-end 3D point trackers such as SpatialTrackerV2." While the authors cite ViPE and SpatialTrackerV2, they do not provide a quantitative comparison, an ablation study, or a reference to a specific benchmark result in the main text or the appendix to substantiate the claim that ViPE is *more accurate*. The text relies solely on the phrase "We empirically find," which is insufficient for a strong comparative claim in a scientific paper. This should be supported by a table in the appendix or rephrased to reflect a qualitative observation without asserting superior accuracy.

2. Overgeneralization of Performance (Abstract, Introduction, Section 4): The abstract and introduction claim that MolmoMotion "significantly outperforms all existing motion prediction baselines." Table 1 (Section 4) presents results for two variants: Autoregressive (AR) and Flow-Matching (FM). While the AR variant generally outperforms baselines, the FM variant on the HOT3D subset shows an ADE of 0.183m compared to ObjectForesight's 0.129m. ObjectForesight is a specialized rigid-object baseline, but the claim that the *model* (implying both variants) outperforms *all* baselines is slightly inaccurate given the FM variant's performance on this specific subset. The claim should be qualified to specify that the autoregressive variant achieves the strongest performance, or the comparison should be restricted to the best-performing variant.

3. Citation Context (Section 1, Introduction): The paper cites 'gibson2014ecological' for the claim that "motion informs an observer how they and other objects move through space, explains object occlusion and permanence, and identifies affordances." While Gibson's work is foundational, the specific phrasing regarding "object occlusion and permanence" is a modern interpretation. The citation is appropriate for the general concept, but the specific list of functions might benefit from a more direct citation to a review paper or a more specific section of Gibson's work if available, though this is a minor point compared to the empirical claims above.

4. Data Availability and Reproducibility: The paper correctly cites external repositories for code and data (HuggingFace, GitHub). The claim that the dataset is "human-verified" is supported by the description of the benchmark construction in Section 3 and Appendix 5, where human verification steps are detailed. No issues found regarding the availability of external resources.

Conclusion: The paper makes strong claims about empirical superiority and performance that are not fully backed by the presented data or citations in the specific instances noted above. The core science is sound, but the precision of the claims requires adjustment to match the evidence provided.

- [science] In Section 3 (Method), the paper claims ViPE produces 'more accurate 3D trajectories than current end-to-end 3D point trackers such as SpatialTrackerV2' based on an 'empirical' finding. However, no quantitative comparison, ablation study, or citation to a specific benchmark result is provided in the text or appendix to substantiate this superiority claim. This assertion requires evidence or rephrasing to 'we observed' without claiming general superiority.

- [writing] The abstract and Introduction claim MolmoMotion 'significantly outperforms all existing motion prediction baselines.' Table 1 shows the AR variant outperforms baselines on most metrics, but the Flow-Matching (FM) variant has higher ADE/FDE than ObjectForesight on the HOT3D subset (0.183 vs 0.129 ADE). The claim of 'significant' outperformance for the *entire* model family is slightly overstated given the FM variant's mixed results on specific subsets compared to rigid-object baselines.

- [writing] Section 4 claims the model 'improves training efficiency' in robotics, citing a jump from 19% to 51% success at 10K steps. While the data supports a faster rise, the claim implies a general efficiency gain. The text does not explicitly rule out that the baseline might eventually reach the same plateau with more steps, though the final gap (56% vs 76%) supports the claim. However, the phrasing 'improves training efficiency' is acceptable but could be more precise as 'accelerates convergence'.

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1
Figure 1 effectively illustrates the pipeline but contains a critical mismatch between the described action ('tan flask') and the visual input (white pitcher), and lacks explicit definition of the time-color mapping in the trajectory panel despite the presence of a colorbar.

Figure 2
The figure provides a clear visual overview of the two model variants (autoregressive and flow-matching) and their inputs. However, the caption contains a citation artifact and omits the 'Action' input from its textual description of the shared inputs.

Figure 3
The figure effectively illustrates the data annotation pipeline steps, but the colorbar at the bottom is unlabeled and the y-axis metric '|□3D|' in the bottom-right graph is undefined, reducing interpretability.

Figure 4
The figure effectively visualizes the distribution of motion verbs in the dataset, but the caption is too brief to explain the chart's content, and the logarithmic y-axis lacks sufficient tick marks for accurate interpretation.

Figure 5
The figure effectively visualizes the long-tail distribution of object categories in the dataset, but the caption is insufficiently descriptive and the logarithmic y-axis lacks necessary gridlines or tick labels to allow for quantitative interpretation of the bar heights.

Figure 6
The figure effectively visualizes diverse motion patterns, but the caption is grammatically incomplete and the legend defining the trajectory colors is not explained in the text.

Figure 7
The figure presents quantitative results for a pick-and-place task but lacks the visual context described in the caption and fails to define the evaluation splits shown in the bar chart.

Figure 8
The figure effectively visualizes predicted 3D trajectories on real-world robotic tasks, but the caption contains a grammatical error (missing subject) and inconsistent capitalization.

Figure 9
The figure presents a visual comparison of video generation results but fails to identify the methods being compared via a legend or labels, and the caption contains a missing subject that obscures the specific claim being made.

Figure 10
The figure fails to substantiate the caption's claim of showing prediction results on DROID clips, as it lacks any quantitative data, axes, or legends required to evaluate the model's performance.

Figure 11

The figure presents a clear visual comparison of video generation models, but the caption is too generic to describe the specific prompts and models shown, and the row labels are poorly formatted.

Figure 12

The figure presents a grid of video frames but is critically missing model labels or a legend to identify the generators, rendering the comparison unintelligible. The caption is also insufficient as it does not specify the prompts or methods shown.

- [science] Figure 1: The 'Predicted Future 3D Point Trajectory' panel shows a colorbar labeled 'Time' with a gradient from light to dark, but the caption does not define what the colors represent or confirm that the gradient maps to time steps, creating ambiguity in interpreting the trajectory evolution.
- [writing] Figure 1: The 'Action' input is described as 'pour water from the tan flask into the red can', but the RGB Observation image shows a white pitcher and a red can — no tan flask is visible, creating a mismatch between the textual instruction and the visual input shown.
- [writing] Figure 2: The caption contains a likely citation artifact ('Molmo2 clark2026molmo2') that should be cleaned up for readability.
- [writing] Figure 2: The 'Action' input block in the diagram is not explicitly mentioned in the caption's list of shared inputs, though it is visually present.
- [writing] Figure 3: The colorbar at the bottom lacks a label or legend entry defining what the color gradient represents (e.g., time, confidence, or velocity), making the visualization ambiguous.
- [writing] Figure 3: The graph in the bottom right corner uses the symbol '|□3D|' on the y-axis without defining it in the caption or axis label, leaving the metric's physical meaning unclear.
- [writing] Figure 4: The caption 'MolmoMotion-1M (pretrain)' is insufficient; it fails to describe the chart's content (distribution of motion verbs) or explain the log-scale y-axis.
- [science] Figure 4: The y-axis is labeled '# Clips (log)' with a single tick at 10^4, but lacks a zero baseline or lower bound, making the relative magnitude of the bars difficult to interpret accurately.
- [writing] Figure 5: The caption 'MolmoMotion-1M (pretrain)' is too generic and does not describe the bar chart's content (object category distribution) or the y-axis metric ('# Clips (log)'); it should explicitly state that the figure shows the frequency distribution of object categories in the pretraining dataset.
- [science] Figure 5: The y-axis is labeled '# Clips (log)' but lacks tick marks or gridlines for values other than 10^4, making it impossible to estimate the magnitude of the bars or the relative differences between categories.

- [writing] Figure 6: The caption 'predicts accurate motion trajectories...' lacks a subject (e.g., 'MolmoMotion') and is grammatically incomplete.
- [writing] Figure 6: The top-left legend labels 'Prediction' and 'GT' are not defined in the caption, leaving the color mapping for the trajectory lines ambiguous.
- [science] Figure 7: The caption describes a 'Pick-and-place task' but the figure displays quantitative performance metrics (Test success rate, Final success rate) without any visual depiction of the task, the environment, or the robot, making it impossible to verify the claim visually.
- [writing] Figure 7: The caption is insufficient as it fails to define the evaluation splits (SS, SU, US, UU) shown on the x-axis of the right panel or explain the specific metrics being compared.
- [writing] Figure 8 caption contains a sentence fragment ('can plan accurate object trajectories...') lacking a subject, likely due to a copy-paste error.
- [writing] Figure 8 caption uses lowercase 'trajectory' in the title line instead of the capitalized 'Trajectory' used in other figure titles (e.g., Figure 2, 7).
- [writing] Figure 9 caption contains a grammatical error and missing subject: '-guided videos exhibit...' should specify the model name (e.g., 'MolmoMotion-guided videos').
- [science] Figure 9 lacks a legend or row labels identifying the competing methods (e.g., 'Wan2.2-I2V-A14B', 'CogVid-5B-I2V', 'DAS – MolmoMotion'), making it impossible to determine which row corresponds to the claimed 'more physically plausible' results.
- [science] Figure 10: The caption claims to show 'predictions on held-out DROID clips,' but the image contains no quantitative data, error bars, or ground-truth comparisons to validate the prediction accuracy.
- [writing] Figure 10: The image consists of nine unlabelled sub-panels with no axes, units, or legends, making it impossible to interpret the specific content or metrics shown.
- [science] Figure 11: The figure displays qualitative video generation results for prompts 'little pigs walks to the big pig' and 'a giant panda holds a bamboo stick and scratches its head', but the caption 'Video generation comparisons on held-out prompts (1/2)' is generic and does not describe the specific content or the models being compared (Wan2.2, CogVid, MolmoMotion).
- [writing] Figure 11: The row labels 'D_A S – MolmoMotion' are vertically compressed and difficult to read; the spacing should be increased for clarity.
- [science] Figure 12: The figure displays qualitative video generation results but lacks any labels, legends, or text identifying which specific model (e.g., MolmoMotion vs. baselines) generated each row or column, making the comparison impossible to interpret.
- [writing] Figure 12: The caption 'Video generation comparisons on held-out prompts (2/2)' is generic and fails to describe the specific prompts used or the models being

compared, which are not visible in the image itself.

paper_reviewer_jargon_police

Verdict: minor_revision
The manuscript relies heavily on specialized terminology that creates a barrier for non-specialist readers. While the technical precision is high, the density of unexplained acronyms and field-specific jargon violates the principle of accessibility.

In the **Abstract**, the term ”flow-matching” is introduced without definition. This is a specific generative modeling technique; replacing it with a plain description like ”modeling trajectory distributions via velocity fields” would improve clarity. Similarly, ”autoregressive” is used to describe the prediction method; a brief clarification (e.g., ”predicting coordinates sequentially, one step at a time”) would help general readers.

In **Section 3 (Method)**, the text assumes familiarity with several acronyms. ”MAD-based outlier criterion” appears in Section 3.1 without defining MAD (Median Absolute Deviation). In Section 3.2, ”RoPE” (Rotary Positional Embeddings) and ”DiT” (Diffusion Transformer) are used without expansion. These are standard in the sub-field but obscure to a broader audience. The term ”egocentric” is used repeatedly (e.g., Introduction, Section 3.1) to describe first-person viewpoints; ”first-person” is a more accessible alternative.

In **Section 4 (Experiments)**, ”6-DoF” is used in the baseline description without spelling out ”six degrees of freedom.”

To meet the standard of inclusive scientific communication, every acronym must be defined at its first occurrence, and field-specific jargon (like ”egocentric” or ”autoregressive”) should be accompanied by a plain-language explanation or synonym. This will ensure the paper’s contributions are accessible to researchers outside the immediate niche of 3D motion forecasting and generative modeling.

- [writing] The manuscript relies heavily on specialized terminology that creates a barrier for non-specialist readers. While the technical precision is high, the density of unexplained acronyms and field-specific jargon violates the principle of accessibility. In the Abstract, the term ”flow-matching” is introduced without definition. This is a specific generative modeling technique; replacing it with a plain description like ”modeling trajectory distributions via velocity fields” would improve clarity. Si

paper_reviewer_logical_consistency

Verdict: minor_revision
The paper presents a coherent narrative for the MolmoMotion framework, but several logical inconsistencies and overgeneralizations weaken the strength of the conclusions drawn from the premises.

First, there is a fundamental tension between the claimed properties of the representation and the defined coordinate system. In the Introduction, the authors argue that

3D points in world coordinates are "view-stable," implying consistency across different camera viewpoints. However, Section 3.1 ("Problem formulation") explicitly defines the world coordinate frame as "anchored at the camera at time t_0." If the camera moves after t_0 (which is common in the unconstrained videos used), the coordinates of static background points or even the object itself (relative to the camera origin) will change. The paper does not clarify whether the "world frame" is actually a global SLAM-based frame or a camera-centric frame. If it is the latter, the claim of "view-stability" is logically unsupported, as the representation is inherently tied to the initial camera pose. This ambiguity undermines the argument that the representation is suitable for transfer across different camera settings.

Second, the conclusion that the model "significantly outperforms all existing motion prediction baselines" (Abstract and Section 4) is an overgeneralization. Table 1 explicitly excludes PointWorld and MotionForcast, stating they are excluded because their models "have not yet been released." While it is standard to exclude unavailable code, claiming to outperform "all" baselines is logically flawed when two specific, closely related works are known to exist but were not evaluated. The conclusion should be qualified to "all *evaluated* baselines" or "all *available* baselines" to maintain logical rigor.

Finally, there is a disconnect between the data annotation pipeline and the model input that requires clarification. The pipeline (Sec 3.1) generates dense trajectories (median 88 points, up to 100) and applies complex filtering based on object-level coherence. However, the model is trained on only $N=8$ query points (Sec 4.1). The paper does not logically explain how the dense, filtered data contributes to the training signal if the model only consumes 8 points. If the filtering is only applied to the 8 sampled points, the "dense" nature of the pipeline is irrelevant to the model's learning. If the filtering uses all 100 points to select the best 8, this selection bias should be described. Without this link, the justification for the complex dense pipeline is logically incomplete relative to the sparse model architecture.

- [science] The claim that 3D points are 'view-stable' (Sec 1) contradicts the methodology in Sec 3.1, which anchors the world frame to the camera at t0. If the camera moves, coordinates change. Clarify if 'view-stable' refers to relative motion or if the frame is truly global (e.g., via SLAM).
- [writing] The claim of outperforming 'all' baselines (Abstract, Sec 4) is logically weakened by excluding PointWorld and MotionForcast from Table 1 due to unreleased code. Qualify the claim to 'all evaluated baselines' to avoid overgeneralization.
- [science] The dense annotation pipeline (100 points) vs. sparse model input (8 points) lacks logical justification. Explain how dense filtering contributes to training if the model only sees 8 points, or if the pipeline is merely for data generation without direct model impact.

paper_reviewer_overreach

Verdict: minor_revision
The paper makes several claims that extend beyond the scope of the provided data and experimental setup.

First, the abstract and introduction state that MolmoMotion ”significantly outperforms all existing motion prediction baselines.” This is an overgeneralization. As shown in Table 1, the baseline ”ObjectForesight” achieves a lower Average Displacement Error (ADE) of 0.129 compared to MolmoMotion-FM (0.183) on the HOT3D subset when using 1 frame of input. While ObjectForesight requires mesh inputs (a limitation the authors note), the absolute claim of outperforming ”all” baselines is factually incorrect without qualifying the specific conditions (e.g., ”outperforms all baselines that do not require explicit mesh inputs” or ”outperforms all baselines in the 3D track category”). The text should be revised to accurately reflect the comparative results in Table 1.

Second, the claim that the model improves ”generalization for robot manipulation” (Abstract) and the conclusion that it ”transfers effectively to robotics planning” (Sec 4.2) are not fully supported by the evidence. The robotics evaluation is restricted to a single simulated task (MolmoSpaces pick-and-place) and a trajectory prediction fine-tuning on DROID videos. The authors explicitly admit in the Limitations section that ”closed-loop real-world robot experiments” are needed. Therefore, claiming broad generalization to ”robot manipulation” implies a robustness across embodiments and real-world dynamics that has not yet been demonstrated. The language should be tempered to reflect that the transfer was observed in *simulated* environments and *offline* video datasets, not in general real-world robotic deployment.

Third, the assertion that the model's trajectories enable the synthesis of videos with ”more realistic object motion” (Abstract, Sec 4.3) conflates metric consistency with physical realism. The evaluation relies on VBench metrics (Subject Consistency, Motion Smoothness, etc.), which measure temporal coherence and visual quality, not physical accuracy. A video can be smooth and consistent but physically impossible. Without a physics-based evaluation or a human study specifically rating ”realism” of motion dynamics, the claim of ”more realistic” motion is an overreach. The authors should clarify that the improvement is in terms of *consistency* and *adherence to the prompt* rather than physical realism.

- [science] The claim that the model 'significantly outperforms all existing motion prediction baselines' (Abstract, Intro) is overreaching. Table 1 shows ObjectForesight achieves lower ADE (0.129) than MolmoMotion-FM (0.183) on HOT3D with 1-frame input. The claim should be qualified to exclude methods with specific input requirements (e.g., mesh inputs) or to specify the exact conditions under which the outperformance holds.
- [science] The assertion that the learned prior 'improves training efficiency and generalization' for robotics (Abstract) overstates the evidence. The robotics results (Sec 4.2) are limited to a single simulated pick-and-place task (MolmoSpaces) and a trajectory-finetuning experiment on DROID. No closed-loop real-robot experiments were conducted (as admitted in Limitations), and the generalization claim is not supported by

diverse real-world robotic benchmarks.

- [science] The claim that predicted trajectories provide 'effective motion guidance' for video generation resulting in 'more realistic object motion' (Abstract, Sec 4.3) relies on VBench metrics which measure consistency and smoothness, not physical realism. The paper does not provide a physical simulation-based evaluation or human study to substantiate the claim of 'more realistic' motion compared to larger baselines like Wan2.2.

paper_reviewer_safety_ethics

Verdict: minor_revision
The manuscript presents a significant contribution to 3D motion forecasting but requires specific clarifications regarding data ethics and safety implications before acceptance.

Data Privacy and Consent: The construction of the MolmoMotion-1M dataset relies on scraping and processing 1.16M unconstrained videos from public sources (Section 3.1, Appendix A.1). While the authors cite the original datasets (e.g., Xperience, HD-EPIC), the paper lacks a specific statement regarding the ethical use of this data for *new* 3D trajectory annotation tasks. Specifically, there is no mention of whether the original licenses permit this specific type of derivative work or if any Personally Identifiable Information (PII) (e.g., faces, license plates) was redacted from the 1.16M clips before training. Given the scale, a statement confirming compliance with original data licenses and the absence of sensitive biometric data in the training set is necessary.

Dual-Use and Physical Safety: The paper demonstrates that the model can predict object trajectories for robot manipulation (Section 4.2) and claims "closed-loop success" in simulation. However, the authors explicitly state in the Conclusion and Appendix that "closed-loop real-robot evaluation" is left to future work. This is a critical safety gap. A model trained on internet videos that predicts physical trajectories could be misused to plan harmful physical interactions if deployed on real hardware without rigorous safety constraints. The paper currently lacks a dedicated discussion on the risks of deploying this technology in the physical world. A "Safety and Limitations" section must be added to explicitly address the potential for physical harm, the necessity of safety layers (e.g., collision avoidance, human-in-the-loop) for real-world deployment, and the limitations of sim-to-real transfer regarding safety.

Content Safety and Bias: The data curation pipeline relies heavily on LLMs (Molmo2, Qwen3) to generate action descriptions and object phrases for the training corpus (Appendix A.1.2). There is no discussion of how the authors ensured the training data does not contain harmful, violent, or unsafe action descriptions (e.g., "break," "throw at," "stab"). Without a filtering mechanism or a statement on the safety of the generated captions, there is a risk of the model learning to predict or facilitate harmful physical actions. The authors should clarify if any safety filtering was applied to the generated text or the source videos.

These issues are primarily writing and disclosure-related but are essential for the ethical deployment of the described technology.

- [writing] The paper aggregates 1.16M videos from public sources (e.g., Xperience, YT-VIS) without explicit mention of consent for 3D motion annotation or downstream model training. Add a statement in Section 3.1 or the Appendix confirming that all source data complies with original licenses and that no personally identifiable information (PII) or sensitive biometric data was retained or used.
- [writing] The robotics transfer section (Sec 4.2) claims improved 'closed-loop success' in simulation but explicitly states 'We leave closed-loop real-robot evaluation to future work.' Given the potential for physical harm in real-world deployment, the paper must include a dedicated 'Safety and Limitations' subsection discussing the risks of deploying this model on physical hardware without further safety validation.
- [writing] The data pipeline uses LLMs (Molmo2, Qwen3) to generate captions and object phrases for 1.16M clips. There is no discussion of potential bias amplification or the inclusion of harmful/unsafe action descriptions in the training corpus. A brief statement on the filtering of unsafe content or the limitations of the automated curation regarding safety is required.

paper_reviewer_scientific_evidence

Verdict: minor_revision
The scientific evidence presented for the core claims of MolmoMotion is generally robust, particularly regarding the scale of the dataset (1.16M clips) and the diversity of the benchmark (111 object categories). The ablation studies in the Appendix (Tab 4) effectively isolate the contribution of the anchor-relative coordinate parameterization and language conditioning, providing strong causal evidence for the architectural choices. The use of human-verified ground truth for the benchmark (Sec 4.2) strengthens the validity of the quantitative comparisons.

However, there are critical gaps in the experimental rigor regarding the evaluation protocol and statistical reporting. First, Section 4.1 states that "we follow best-of-5 evaluation for each sample" for all methods. It is ambiguous whether the proposed MolmoMotion models (both AR and FM) were evaluated using this sampling strategy or a single deterministic pass. If the baselines (e.g., pixel-space generators) were evaluated with best-of-5 selection while MolmoMotion was not, the reported superiority in ADE/FDE is artificially inflated. The authors must explicitly confirm the sampling strategy for every row in Table 1.

Second, the claim that pixel-space video generators fail at metric 3D prediction (Table 1) relies on a specific post-hoc pipeline (ViPE depth estimation – AllTracker) to lift 2D video outputs to 3D. The paper does not quantify the error floor of this lifting pipeline itself. If the lifting pipeline introduces significant noise or bias, the poor performance of Wan2.2 and Cosmos may reflect the limitations of the evaluation metric rather than the video generators' inability to model motion. A control experiment measuring the lifting pipeline's error on ground-truth video frames is necessary to validate the comparison.

Finally, the robotics transfer results (Figure 3a) present mean success rates across four splits but omit variance metrics (standard deviation) and the number of random seeds used for training the downstream policies. Given the stochastic nature of policy optimization and the relatively small sample size of the MolmoSpaces benchmark, the statistical significance of the ”substantial improvement” (e.g., 51% vs 19%) cannot be assessed without this information. The authors should report standard deviations or confidence intervals to support the claim of improved generalization.

- [science] The evaluation protocol relies on 'best-of-5' sampling for all baselines (Sec 4.1), but the text does not explicitly state if the proposed MolmoMotion models also used best-of-5 or a single deterministic pass. If MolmoMotion used a single pass while baselines used sampling, the reported ADE/FDE improvements are inflated. Clarify the sampling strategy for all methods in Tab 1.
- [science] The claim that pixel-space methods (Wan2.2, Cosmos) perform poorly on metric 3D tasks (Tab 1) relies on a post-hoc lifting pipeline (Sec 4.1 Baselines). The error introduced by this specific lifting pipeline (ViPE – AllTracker) is not quantified separately from the video generator's failure. A control experiment showing the lifting pipeline's error on ground-truth video frames is needed to isolate the generator's contribution to the metric error.
- [science] The robotics transfer results (Fig 3a) show a large gap in success rates (51% vs 19% at 10K steps). The text attributes this to the motion prior, but does not report the variance (standard deviation) across the 4 splits (SS, SU, US, All) or the number of seeds used for the policy training. Without variance estimates, the statistical significance of the 'substantial improvement' claim is unclear.

paper_reviewer_statistical_analysis

Verdict: minor_revision
The statistical analysis presented in the paper is generally sound in its choice of metrics (ADE, FDE, PWT) and evaluation protocols (best-of-5, held-out benchmarks). However, the reporting of results lacks necessary statistical rigor regarding uncertainty quantification.

In Section 4.1 (Table 1), the authors report single-point estimates for ADE, FDE, and PWT across three datasets (HOT3D, WorldTrack, DAVIS). While the improvements over baselines appear significant in magnitude, the absence of confidence intervals, standard deviations, or standard errors makes it impossible to determine if these differences are statistically significant or merely due to variance in the specific test set composition. For a benchmark of 742 clips, reporting the mean and standard deviation of the error metrics across clips is standard practice.

Furthermore, the ”best-of-5” evaluation strategy described in Section 4.1 introduces a selection bias that must be accounted for. The text states, ”We follow the point-track forecasting metrics... using best-of-5 evaluation.” It is unclear if the reported ADE/FDE values are the average error of the best sample, or if the metric is computed on the single best sample. If the latter, the variance of the selection process should be reported. Without this, the comparison against baselines (which may or may not use the same selection strategy) is potentially confounded.

In Section 4.2, the robotics transfer results (Fig. 4a) show success rates over training steps. The curves are presented without error bands (e.g., 95% confidence intervals over multiple seeds or episodes). Given the high variance often observed in robot learning experiments, the claim that the model "substantially improves" performance requires statistical validation (e.g., a t-test or non-parametric equivalent) to rule out random fluctuation, especially at the 10K step mark where the gap is 51% vs 19%.

Finally, the video generation metrics in Table 2 (VBench scores) are reported to three decimal places (e.g., 0.968). This level of precision implies a high degree of certainty that is likely unsupported by the sample size or the inherent noise in perceptual metrics. Reporting these with appropriate significant figures or confidence intervals would be more scientifically honest.

The paper would benefit from a revised results section that includes measures of dispersion (std dev, CI) for all primary quantitative claims and explicit statistical testing for the key comparative results.

- [science] Report confidence intervals or standard deviations for all quantitative metrics in Table 1 (motion prediction) and Table 2 (video generation). The current presentation of single-point estimates (e.g., ADE=0.109) lacks statistical context regarding variance across the 742 benchmark clips, making it difficult to assess the robustness of the reported improvements over baselines.
- [science] Clarify the statistical methodology for the 'best-of-5' evaluation mentioned in Sec 4.1. Specify whether the final metric is the mean of the best-of-5 samples or the best-of-5 mean, and provide the variance across these selection events to ensure the reported gains are not artifacts of selection bias.
- [science] In the robotics transfer section (Sec 4.2), the success rates (e.g., 76.3% vs 56.0%) are presented without error bars or significance testing. Given the stochastic nature of robot policies and the finite number of evaluation episodes, a statistical test (e.g., bootstrap confidence intervals or a t-test) is required to substantiate the claim of 'substantial improvement'.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript is generally well-written, with a clear narrative flow and precise technical descriptions. The abstract effectively summarizes the contributions, and the introduction successfully motivates the problem. However, there are several minor grammatical errors and typos that should be corrected to ensure professional polish.

Specifically, in the Abstract, the sentence "MolmoMotion is able to accurately predicts" contains a subject-verb agreement error; "predicts" should be "predict". In Section 3, under the description of the training stages, the word ”varients” is misspelled and should be ”variants”. Additionally, in the Conclusion, the opening sentence of the Limitations section (”This work would not be possible...”) uses the wrong tense; it should read ”would not have been possible” to reflect the completed nature of the work.

While the text is largely clear, some sentences in the Introduction and Experiments sections are slightly dense. For instance, the comparison of model sizes in the Introduction could be more explicit. Overall, the writing quality is high, but these specific corrections are necessary before final publication.

- [writing] In the Abstract, correct the subject-verb agreement error: ’MolmoMotion is able to accurately predicts’ should be ’accurately predict’.
- [writing] In Section 3 (Method), fix the typo ’varients’ to ’variants’ in the paragraph describing the second training stage.
- [writing] In Section 4 (Experiments), the phrase ’much larger image-to-video models’ in the Introduction summary is vague; consider specifying the parameter counts or model names for clarity.
- [writing] In the Conclusion, the sentence ’This work would not be possible without...’ contains a grammatical error: ’would not be possible’ should be ’would not have been possible’ to match the past tense context.